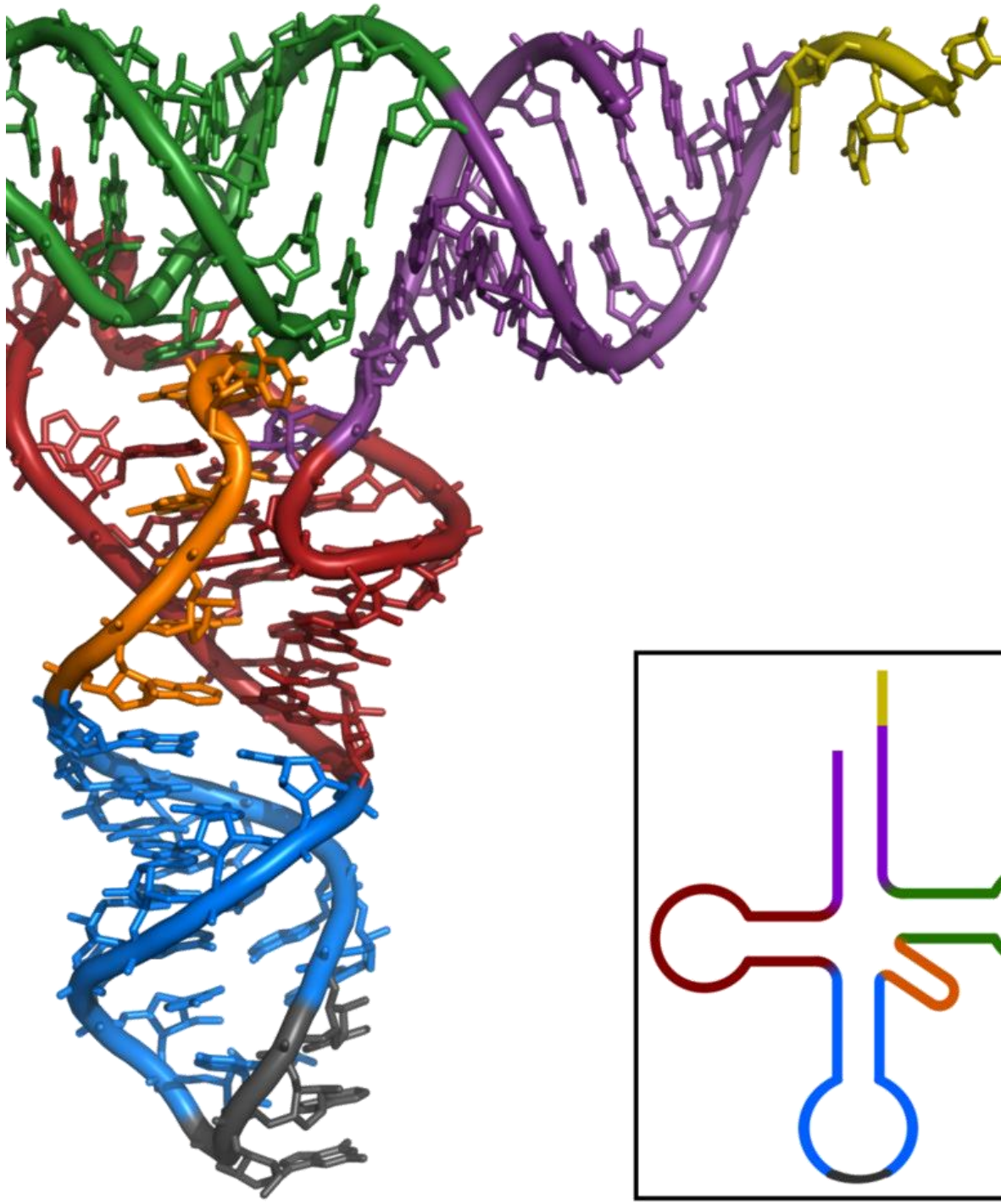
DP Biology – D1.2

# Protein synthesis

- How does a cell turn a DNA base sequence into a chain of amino acids?
- What keeps protein synthesis reliable?

SL statements D1.2.1–11. Textbook pp. 615–624.
Image: model of a transfer RNA molecule. Yikrazuul, CC BY-SA 3.0.



# Objectives for each class

| Class | Statements | By the end of the class, you can... |
|---|---|---|
| **1** | D1.2.1–4 | Describe transcription, and explain why DNA templates stay stable and how transcription controls which genes are expressed |
| **2** | D1.2.5–9 | Outline translation, explain the features of the genetic code, and use a table of codons |
| **3** | D1.2.10–11 | Describe how the ribosome builds a polypeptide, and explain how one base change alters a protein |

# From gene to protein

D1.2.1




A gene's **base sequence** sets the **amino acid sequence** of a polypeptide. **Transcription** copies the gene into mRNA; **translation** uses the mRNA to build the polypeptide.

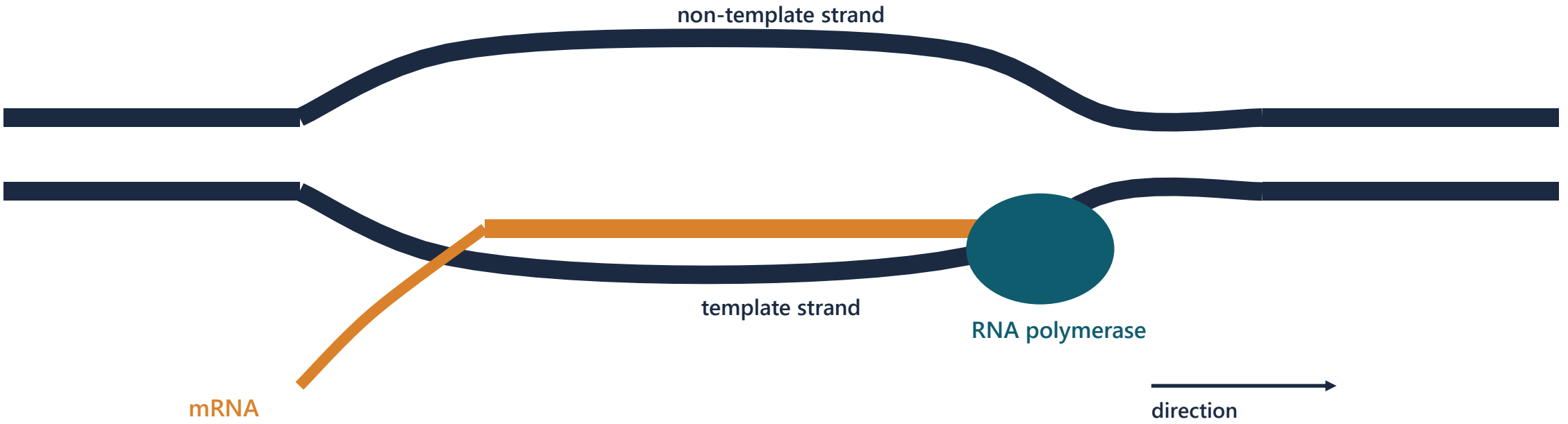
Proteins then shape most of an organism's characteristics.

**Common mistake**

Transcription keeps the language of bases. Translation changes it, from bases into amino acids.

# Transcription and RNA polymerase

D1.2.1




**RNA polymerase** separates the DNA strands and pairs RNA nucleotides with the **template strand**, linking them with covalent bonds. The mRNA peels away and the helix closes behind.

Drawn, not to scale. One enzyme does it all: no helicase or DNA polymerase.

# Base pairing in transcription

D1.2.2

| Strand | Base sequence |
|---|---|
| **DNA non-template strand** | A T G C C A G T |
| **DNA template strand** | T A C G G T C A |
| **mRNA** | A U G C C A G U |

Complementary bases form **hydrogen bonds**. RNA has **uracil** instead of thymine, so A on the template pairs with U.

So mRNA matches the non-template strand, with U in place of T.

**Try it**

A non-template strand reads A G G T T C A G T. Write the template strand, then the mRNA.

# Why DNA templates stay stable

D1.2.3

Transcription copies a sequence **without changing the DNA**, so a gene can be transcribed again and again.

Neurons do not divide, so they keep the same DNA for life: its sequence must stay unchanged for decades.

### Why DNA is stable

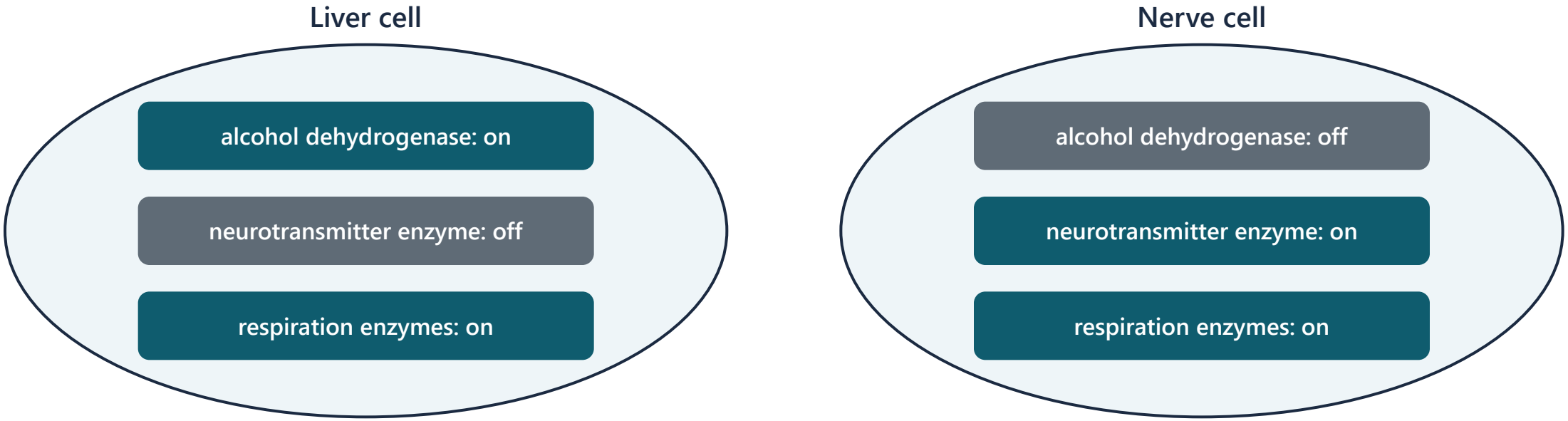
Bases are tucked inside the helix and held by hydrogen bonds; the sugar-phosphate backbone is held by strong covalent bonds.

### Link

Radiation or chemicals can still damage bases. Repair enzymes fix most damage; what remains is mutation (D1.3).

# Transcription controls gene expression

D1.2.4




**Gene expression** turns a gene's information into an effect. Transcription is the first step, so it is the main point where genes are switched **on or off**.

**Common mistake**

Different cells in one body have the same genes. They differ in which genes are transcribed.

# Practice: transcribe

D1.2.1–4

| Strand | Sequence |
| --- | --- |
| **DNA template** | T A C C G A T T G |
| **mRNA** | ? |
| **DNA non-template** | ? |

1. Complete the table.
2. How many codons are in this mRNA?
3. Explain why a liver cell and a skin cell make different proteins.

# Check-in: class 1

D1.2.1–4

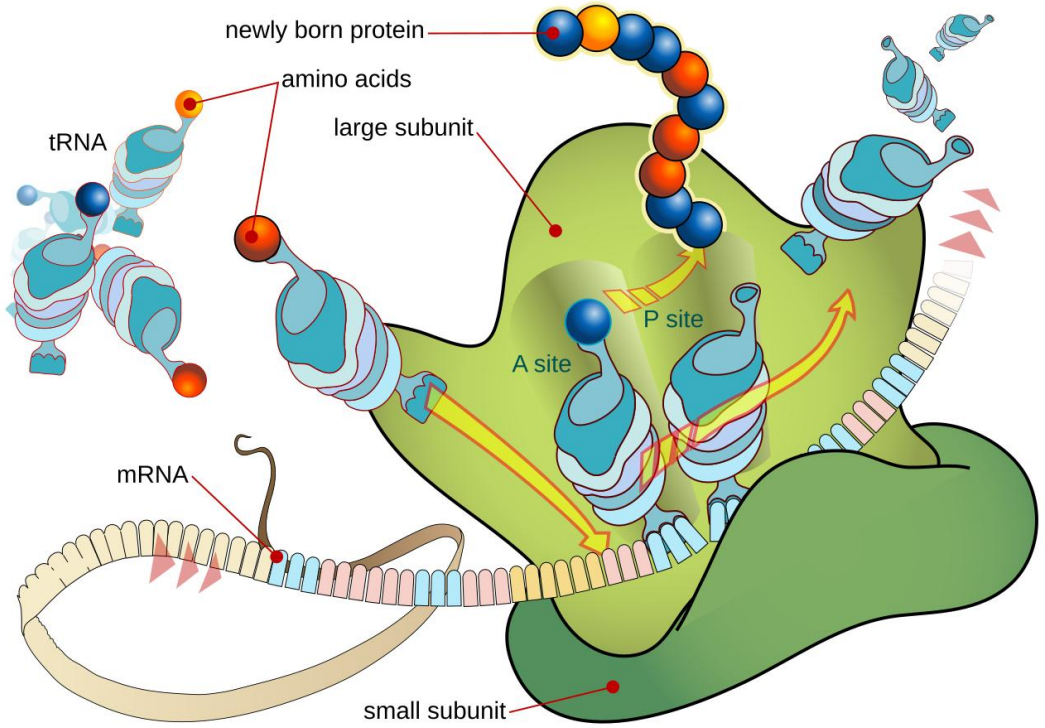
1. State the role of RNA polymerase.
2. Explain how base pairing gives the correct mRNA.
3. State one difference between DNA and RNA bases.
4. Explain why DNA templates can be used many times.
5. Explain how transcription controls gene expression.

**How to use this**

Answer without your notes, then check against slides 3–8.

# Translation: three types of molecule

D1.2.5–6




A ribosome translating mRNA. LadyofHats, public domain.

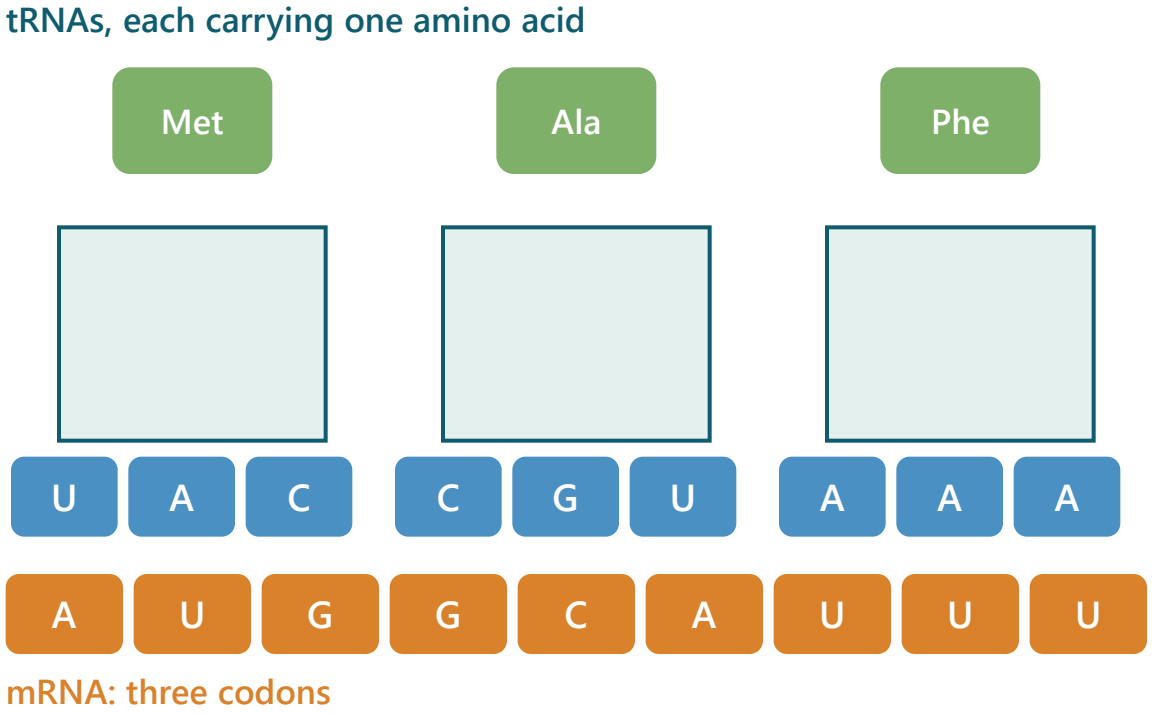
| Molecule | Role |
|---|---|
| **mRNA** | Carries codons; binds the small subunit |
| **Ribosome** | Holds mRNA and tRNA; makes peptide bonds |
| **tRNA** | Brings an amino acid; two fit the large subunit at once |

**Key idea**

**Translation** builds a polypeptide whose amino acid order is set by the mRNA base sequence.

# Codons and anticodons






A **codon** is three bases on mRNA; an **anticodon** is three bases on tRNA.

They pair by complementary bases, so each codon brings in the matching amino acid.

# Features of the genetic code

D1.2.8

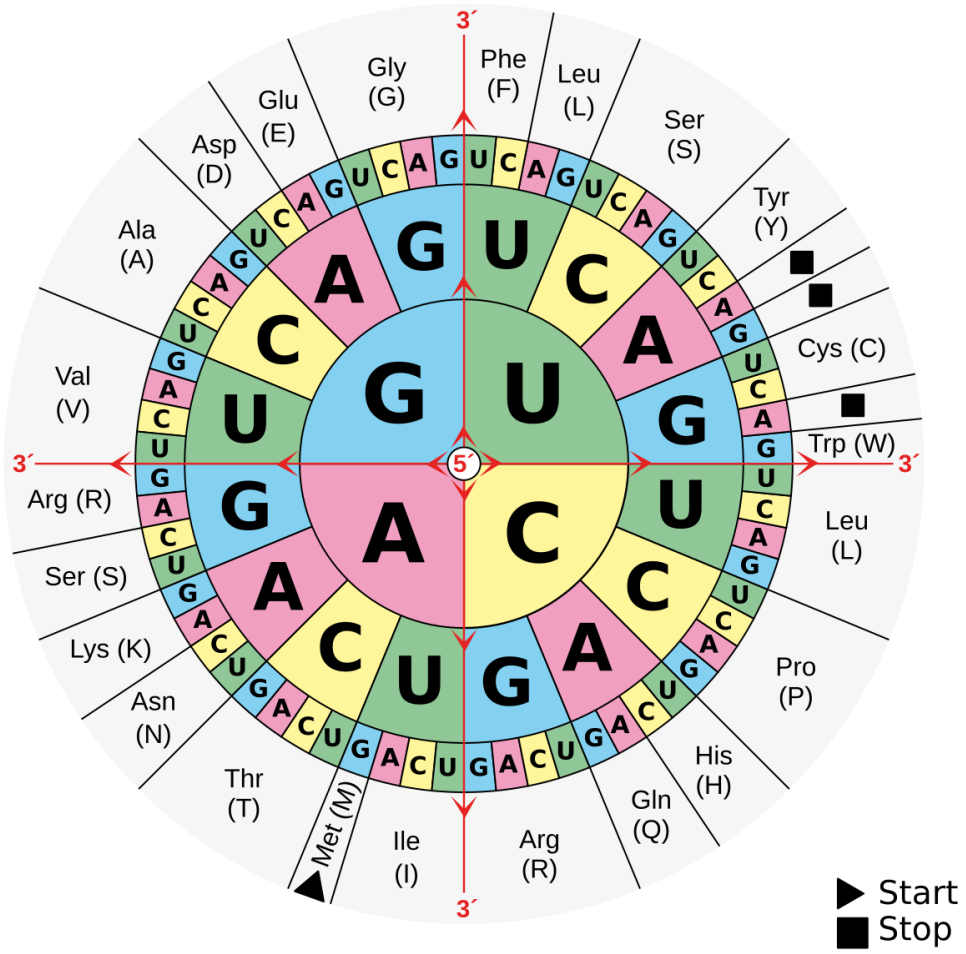
| Feature | What it means |
|---|---|
| **Triplet** | 3 bases per amino acid. 2 bases give only 16 combinations; 3 give 64, enough for 20 |
| **Degenerate** | Several codons can give the same amino acid: GCU, GCC, GCA, GCG = alanine |
| **Universal** | Almost all organisms read codons the same way: evidence of common ancestry |
| **Start and stop** | AUG starts; three codons mean stop |

**Common mistake**

Degenerate: many codons, one amino acid. Never one codon giving several amino acids.

# Reading a table of mRNA codons






Amino acids by mRNA codon, read from the centre outwards. Mouagip, public domain.

| mRNA | AUG \| GCU \| UGU \| UAA |
|---|---|
| **Polypeptide** | Met – Ala – Cys – stop |

Split the sequence into codons from the **first base**. Look up each in turn. A stop codon adds no amino acid.

### Try it

Decode C A C G G G. Then write the DNA template strand for the same mRNA.

# Practice: decoding a gene

D1.2.9

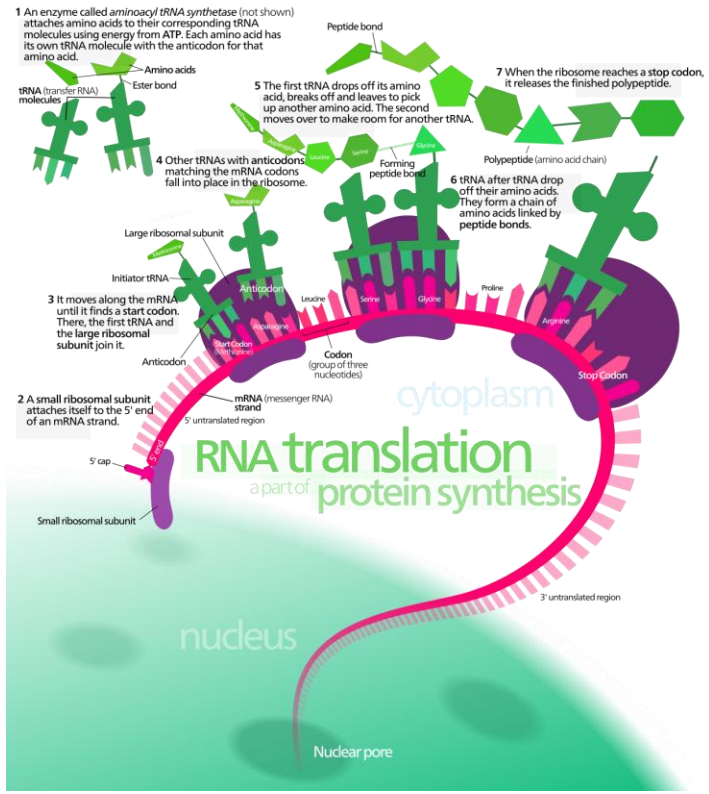
Part of the non-template strand of a gene. Use the codon table on slide 13.

| Non-template strand (DNA) | ATG TTT GGC AAA TGA |
|---|---|
| **mRNA** | ? |
| **Template strand (DNA)** | ? |
| **Amino acids** | ? |

1. Complete the table.
2. A mutation changes TTT to TTC. Predict the effect.
3. A mutation changes AAA to TAA. Predict the effect.

# Why a universal code matters

D1.2.8




Protein synthesis overview. CC BY 3.0.

Because almost every organism reads codons the same way, a human gene put into a bacterium is translated into the **same** protein.

That is how bacteria make human insulin for people with diabetes.

### Think

What does a shared code suggest about the history of life?

# Check-in: class 2

D1.2.5–9

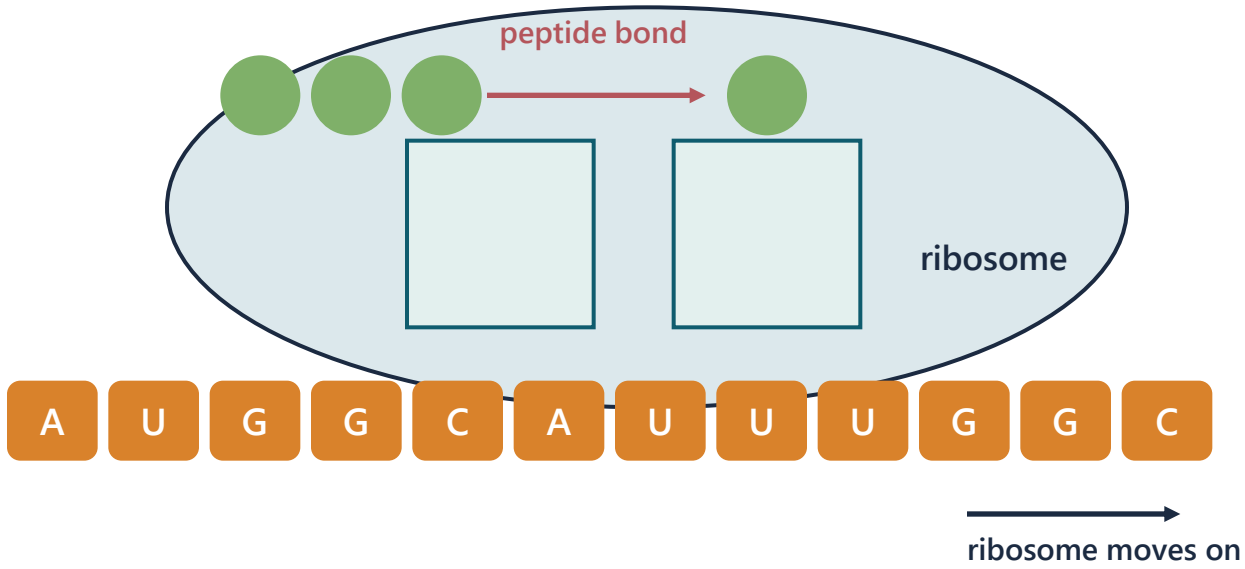
1. State the roles of mRNA, tRNA and ribosomes.
2. Distinguish between a codon and an anticodon.
3. Explain why the code must be a triplet code.
4. Explain what degenerate means, with an example.
5. Translate the mRNA AUG GCU UAC UGA.

**How to use this**

Answer without your notes, then check against slides 10–15.

# Building the polypeptide






tRNA pairs with the next codon → Peptide bond joins its amino acid → Ribosome moves one codon → Empty tRNA leaves

# Transcription and translation compared

D1.2.10

| | Transcription | Translation |
|---|---|---|
| **Where** | Nucleus (eukaryotes) | Ribosomes in cytoplasm or on rough ER |
| **Reads** | DNA template strand | mRNA codons |
| **Product** | mRNA | Polypeptide |
| **Molecules** | RNA polymerase, RNA nucleotides | mRNA, tRNA, ribosome, amino acids |
| **Bonds made** | Between nucleotides | Peptide bonds |

# One base change, one amino acid change

D1.2.11

| | Normal allele | Sickle cell allele |
|---|---|---|
| **DNA non-template** | G A G | G T G |
| **mRNA codon** | G A G | G U G |
| **Amino acid** | Glutamic acid | Valine |

One base is **substituted** in the gene for beta-globin (part of haemoglobin), so one amino acid changes.

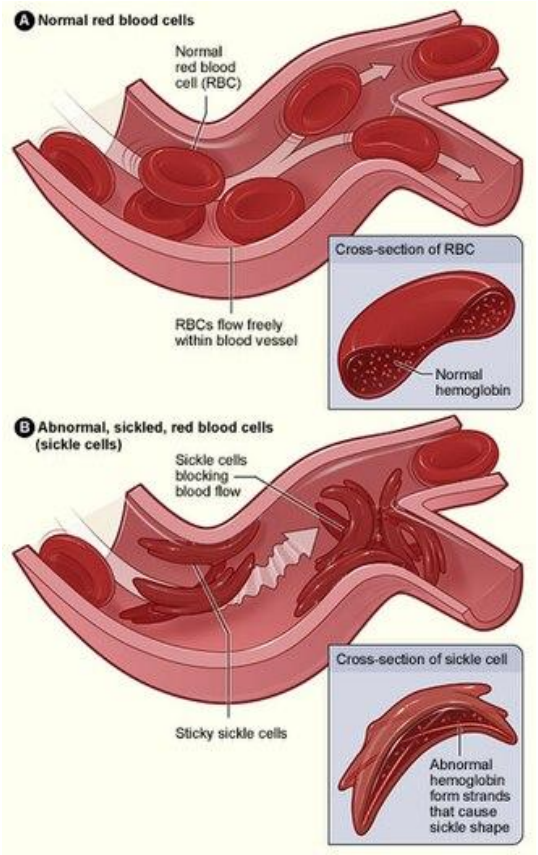
Valine is non-polar, glutamic acid is charged: the protein folds and behaves differently.

**Key idea**

Structure depends on sequence, so one base can affect the whole organism.

# Sickle cell anaemia






Normal and sickled cells. NHLBI, public domain.

At low oxygen the altered haemoglobin molecules stick together into fibres. Red cells become **rigid** and **sickle-shaped**.

They block small vessels, causing pain, and break down early, causing **anaemia**.

**Link**

You will meet this allele again in D1.3 and D3.2.

# Practice: predict the effect

D1.2.11

| Change | Original codon | New codon |
|---|---|---|
| **A** | GAA (Glu) | GAG |
| **B** | UGG (Trp) | UGA |
| **C** | CCU (Pro) | CAU |

1. Use the codon table to find each new amino acid.
2. Which change has no effect?
3. Which makes a shorter protein?

# Working with data: oxygen and sickling

D1.2.11

Percentage of red cells sickled in blood from a person with sickle cell anaemia. The values are invented for practice; the pattern is realistic.

| Oxygen / kPa | Sickled cells / % |
|---|---|
| 13 | 2 |
| 8 | 10 |
| 5 | 35 |
| 3 | 70 |

1. Describe the trend.
2. Explain why sickling is worse in muscles during exercise.
3. Suggest why the person may feel pain at high altitude.

# Check-in: class 3

D1.2.10–11

1. Describe one cycle of elongation in translation.
2. Distinguish between transcription and translation.
3. State the base change in the sickle cell allele.
4. Explain how this changes the protein.
5. Explain why the cells sickle at low oxygen.

**How to use this**

Answer without your notes, then check against slides 17–22.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **mRNA is made in the cytoplasm.** | Transcription happens in the nucleus. |
| **tRNA carries the code from the nucleus.** | mRNA carries the code; tRNA brings amino acids. |
| **The anticodon is on mRNA.** | Codon on mRNA; anticodon on tRNA. |
| **One gene codes for one amino acid.** | One codon codes for one amino acid. |
| **RNA contains thymine.** | RNA has uracil instead. |

# Summary: protein synthesis

D1.2.1–11

| | Key idea |
|---|---|
| **D1.2.1–2** | RNA polymerase builds mRNA on the template strand; A pairs with U |
| **D1.2.3–4** | DNA unchanged by transcription; transcription switches genes on |
| **D1.2.5–7** | mRNA, tRNA and ribosome; codon pairs with anticodon |
| **D1.2.8–9** | Triplet, degenerate, universal; use the codon table |
| **D1.2.10** | tRNA in, peptide bond, ribosome moves |
| **D1.2.11** | GAG to GTG: valine for glutamic acid, sickle cells |

# Check that you can...

1. **Outline** transcription and the role of RNA polymerase
2. **Explain** how base pairing gives the right sequence
3. **Describe** translation and the roles of mRNA, tRNA and ribosomes
4. **Use** a codon table to find amino acid sequences
5. **Explain** how a base substitution can change a protein

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. A DNA template strand reads T A C G G A. Write the mRNA.
2. Give the anticodon that pairs with the codon G C A.
3. Explain what is meant by saying that the genetic code is degenerate.

# Image credits

TRNA-Phe yeast 1ehz.png – Yikrazuul – CC BY-SA 3.0
Ribosome mRNA translation en.svg – LadyofHats – Public domain
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